

Chapter 40

MIPI CSI

40.1 Introduction

The MIPI CSI module of ESP32-P4 provides a standard interface with camera modules for mobile devices, such as smartphones, tablets, and other portable electronics. It connects cameras to application processors in mobile devices, enabling high-speed data transfer and efficient communication between the camera and the host system.

ESP32-P4 MIPI CSI implements a MIPI RX D-PHY and a MIPI CSI-2 host controller. It provides one clock lane and two data lanes, supporting a data transmission of up to 1.5 Gbps per lane to communicate with camera sensors compliant with the [MIPI CSI-2 specification](#).

Note:

The [MIPI CSI-2 specification](#) defines the communication between image application processors and cameras. It is part of the communication protocols defined by MIPI Alliance standards, designed for chip-to-chip communication in mobile systems.

40.2 Terminology

This section covers terminology used to describe the functionality of MIPI CSI.

MIPI CSI-2	MIPI Alliance Specification for Camera Serial Interface 2. It receives camera data through the PHY Protocol Interface (PPI) from the MIPI RX D-PHY and outputs data through a 32-bit image interface to the Image Signal Processor (ISP).
MIPI D-PHY	MIPI Alliance Specification for the source synchronous physical layer. It implements the physical layer of universal lanes for the MIPI D-PHY interface, which receives clock and data from MIPI camera sensors through one clock lane and two data lanes.
PPI	PHY Protocol Interface defined in MIPI Alliance Specification. It connects the CSI-2 host controller and D-PHY.
ISP	Image Signal Processor, used to process images received from the camera. For more details about ISP, please refer to Chapter 36 Image Signal Processor (ISP) .
Transmitter	The device that transmits data, usually a MIPI camera.
Receiver	The device that receives data, in this chapter, the ESP32-P4 chip.
Clock lane	The clock lane of the MIPI D-PHY interface, used to transmit high-speed differential clock signals.

Data lane	The data lane of the MIPI D-PHY interface, used to transmit data and commands.
Line	The line of the data lane of the MIPI D-PHY interface; each data lane consists of two lines: P and N.
Sequence	A string of state codes received through the data lane, controlling the MIPI RX D-PHY to enter other modes from Control mode.
Control mode	One of the operating modes of the MIPI RX D-PHY, used to receive requests from the transmitter. All other modes return to this mode.
High-Speed Data Reception mode	One of the operating modes of the MIPI RX D-PHY, used to receive data in burst mode.
Escape mode	One of the operating modes of the MIPI RX D-PHY, used to support low-speed asynchronous communication on the data lane to maintain a low-power state.
Ultra Low Power State (ULPS)	One of the operating modes of the MIPI RX D-PHY, in which the clock lane and data lane are in idle state, maintaining the lowest power consumption except for the Shut-Down state.
Test code	Specific control or configuration codes used to manage and test the MIPI D-PHY, such as setting operational mode.

40.3 Feature List

MIPI CSI in ESP32-P4 supports the following features:

- MIPI RX D-PHY compliant with MIPI D-PHY interface specification revision 1.1
- MIPI CSI-2 host compliant with MIPI CSI-2 specification
- Two data lanes, each with a rate ranging from 80 Mbps to 1.5 Gbps
- Ultra Low Power State in Escape mode
- 32-bit image interface with ISP
- Various input image formats:
 - RGB888/RGB666/RGB565
 - YUV422/YUV420
 - RAW8/RAW10/RAW12
- Error detection and correction at PHY level, packet level, line level, and frame level
- Electromagnetic Interference (EMI) mitigation by data scrambling

Note:

ESP32-P4 does not support different virtual channels defined in the MIPI specification, so virtual channel numbers in the packet header are ignored.

40.4 Architectural Overview

Figure 40.4-1 shows the architecture of MIPI CSI and its connection to other system components. The MIPI CSI consists of the following components:

- **MIPI RX D-PHY:** Implements the physical link layer, including one clock lane (CP/CN) and two data lanes (D0P/D0N and D1P/D1N) to connect with cameras.
- **MIPI CSI-2 host:** Implements the CSI-2 protocol, receives camera data from D-PHY, and outputs through the Image Interface, which includes:
 - **Pipeline:** Registers PPI signals.
 - **Descrambler:** Converts PPI scrambled data to its original values when enabled.
 - **PHY adaptation layer:** Manages the PHY interface, including PHY error handling.
 - **Packet analyzer:** Processes received data from lanes, including data lane merging, header decoding, and various error detection and correction.
 - **Image interface:** Reorders pixels into 32-bit Image Interface data and generates timing-accurate video synchronization signals.
 - **Error management:** Monitors and notifies about error conditions on the CSI-2 link.
 - **Register bank:** Provides access to configuration and control registers.
- **PPI interface:** Connects the MIPI RX D-PHY and the MIPI CSI-2 host controller.
- **Test code interface:** Sends configuration details from the MIPI CSI-2 host controller to the MIPI RX D-PHY.

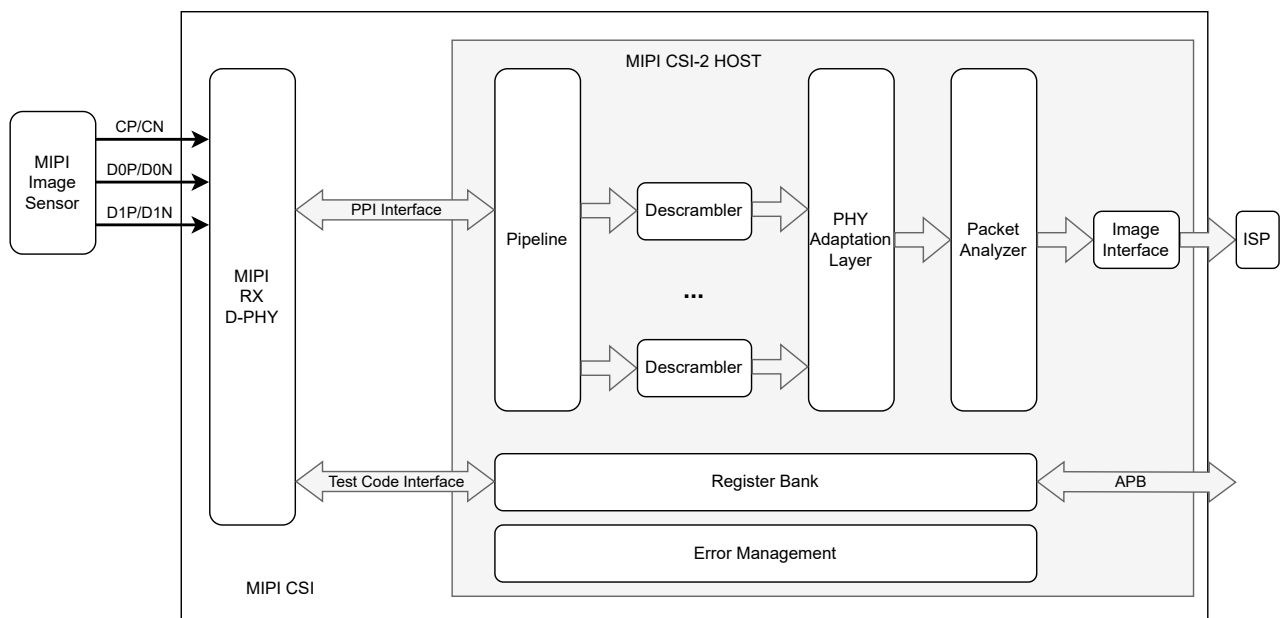


Figure 40.4-1. MIPI CSI Architecture

40.5 Functional Description

This section introduces the operating states and modes of the MIPI RX D-PHY and the major functions that facilitate the use of MIPI CSI. For detailed configurations, refer to Section [40.7 Programming Procedures](#).

40.5.1 Data Lane State

The functions of the transmitter determine the data lane state by driving certain line levels. During normal operation, either an HS-TX or an LP-TX is driving a lane.

- HS-TX: High-Speed transaction. An HS-TX always drives the lane differentially. It operates in two states: HS-0 and HS-1. When HS-TX drives the lane, the lane remains in High-Speed Data Reception mode.
 - HS-0: Transmits differential data 0 in High-Speed Data Reception mode.
 - HS-1: Transmits differential data 1 in High-Speed Data Reception mode.
- LP-TX: Low-Power transaction. The two LP-TXs drive the two lines of a lane independently and single-ended. It operates in four states: LP-00, LP-01, LP-10, and LP-11. When LP-TX drives the lane, the lane is in Control mode or Escape mode.
 - LP-00: A bridge state between other LP states (LP-01, LP-10, LP-11) in Control Mode or Escape Mode. When in Ultra Low Power State (ULPS), lanes are also in the LP-00 state.
 - LP-01: Transmits data 0 in Escape mode, or indicates entry into High-Speed Data Reception mode from Control mode.
The complete sequence for entering High-Speed Data Reception mode is: LP-11 → LP-01 → LP-00
 - LP-10: Transmits data 1 in Escape mode, or indicates entry into Escape mode or Turnaround mode from Control mode.
The complete sequence for entering Escape mode is: LP-11 → LP-10 → LP-00 → LP-01 → LP-00.
The complete sequence for entering the Turnaround mode is: LP-11 → LP-10 → LP-00 → LP-10 → LP-00
 - LP-11: The stop state, i.e., the default idle state of a lane.

Note:

The Turnaround mode is used for switching the data transmission direction. This mode is not used in the ESP32-P4 MIPI CSI, and will not be further explained in this chapter.

Figure [40.5-1](#) shows the lane line levels for High-Speed (HS) and Low-Power (LP) transactions.

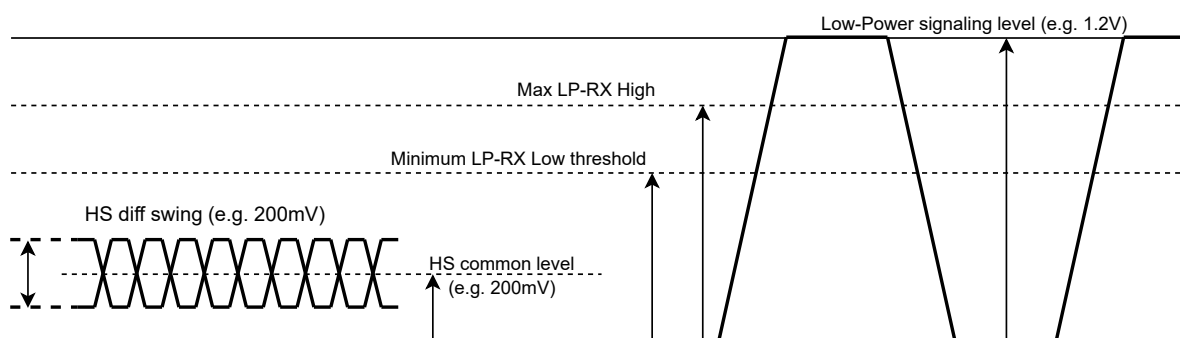


Figure 40.5-1. Level of MIPI Data Lane Lines

Table 40.5-1 shows the available line states.

Table 40.5-1. Lane State Descriptions

State Code	DP Line	DN Line	Corresponding State in Different Modes		
			High-Speed Data Reception Mode	Control Mode	Escape Mode
HS-0	HS Low	HS High	0	N/A	N/A
HS-1	HS High	HS Low	1	N/A	N/A
LP-00	LP Low	LP Low	N/A	Bridge	Space
LP-01	LP Low	LP High	N/A	HS-Rqst	0
LP-10	LP High	LP Low	N/A	LP-Rqst	1
LP-11	LP High	LP High	N/A	Stop	N/A

40.5.2 MIPI RX D-PHY Operation

Figure 40.5-2 illustrates the various states and modes of the MIPI RX D-PHY during initialization and active state. Detailed descriptions of each states and modes are presented in the following sections.

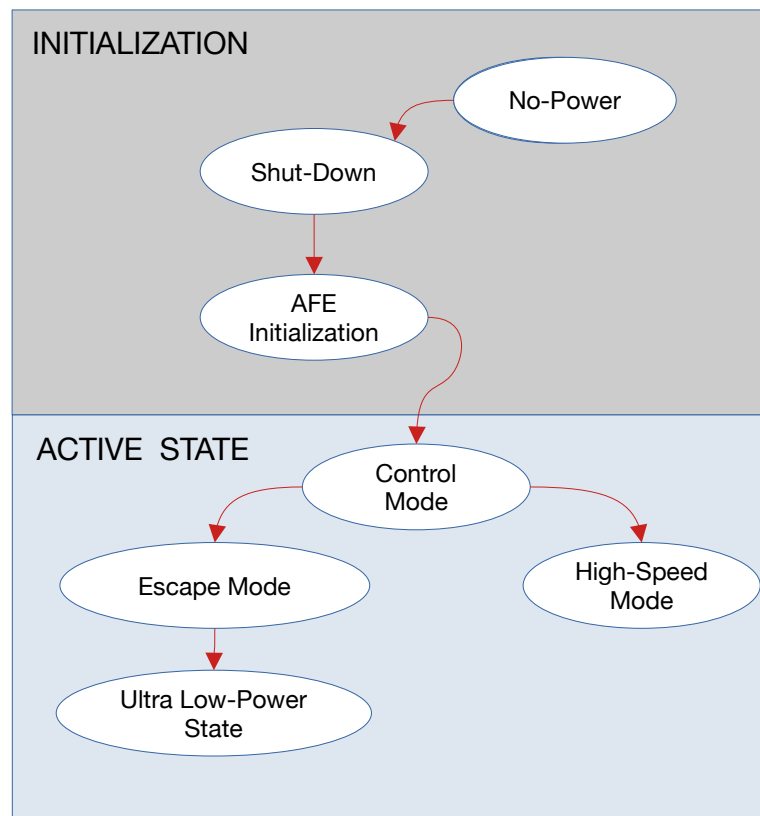


Figure 40.5-2. MIPI RX D-PHY Operating States and Modes

40.5.2.1 No-Power State

In the No-Power state, no supply voltage is applied to the PHY.

The MIPI RX D-PHY of MIPI CSI is in No-Power state when the power supply `VDD_HP_n` (n: 0-3) and `VDD_MIPI_DPHY` are off.

40.5.2.2 Shut-Down State

Shut-Down state refers to the idle state.

In Shut-Down state, the PHY disables all analog blocks, resets all digital logic, and sets the DOP/DON, D1P/D1N, and CP/CN lanes to high impedance (Hi-Z). Activating the Shut-Down state ensures the lowest power consumption. To enable the Shut-Down state of the PHY, set `CSI_HOST_DPHY_RSTZ` and `CSI_HOST_PHY_SHUTDOWNZ` to 0.

Before leaving the Shut-Down state, configure several parameters, including the lane number to be used and the lane operation frequency range.

The MIPI RX D-PHY consists of two data lanes. Configure `CSI_HOST_N_LANES` to select either one or both data lanes. These configurations should remain static or stable before exiting the Shut-Down state.

Depending on the specific requirements for the MIPI RX D-PHY, additional configuration steps might be necessary. By default, the PHY operates within a lower range of 80-110 Mbps. To accommodate higher bit rates, adjust the HS frequency ranges (the `hsfreorange` field) in test code `HS RX Control of lane 0` with the appropriate code. It is recommended to perform this adjustment in the Shut-Down state, as the control interface is independent of the rest of the MIPI RX D-PHY.

The different ranges supported are shown in Table 40.5-2. The hsfreqrange field indicates the HS operating frequency range and is accessible through test code 0x44. For detailed configuration, please refer to Section 40.5.3 *MIPI RX D-PHY Configuration*.

Table 40.5-2. Frequency Ranges Supported by MIPI RX D-PHY

Frequency Range (Mbps)	hsfreqrange[5:0]
80-89	000000
90-99	010000
100-109	100000
110-129	000001
130-139	010001
140-149	100001
150-169	000010
170-179	010010
180-199	100010
200-219	000011
220-239	010011
240-249	100011
250-269	000100
270-299	010100
300-329	000101
330-359	010101
360-399	100101
400-449	000110
450-499	010110
500-549	000111
550-599	010111
600-649	001000
650-699	011000
700-749	001001
750-799	011001
800-849	101001
850-899	111001
900-949	001010
950-999	011010
1000-1049	101010
1050-1099	111010
1100-1149	001011
1150-1199	011011
1200-1249	101011
1250-1299	111011
1300-1349	001100
1350-1399	011100
1400-1449	101100

Frequency Range (Mbps)	hsfreqrange[5:0]
1450-1500	111100

When [CSI_HOST_DPHY_RSTZ](#) and [CSI_HOST_PHY_SHUTDOWNZ](#) are set to 1, the MIPI RX D-PHY leaves the Shut-Down state and starts the initialization procedure.

40.5.2.3 AFE Initialization

After releasing [CSI_HOST_DPHY_RSTZ](#) and [CSI_HOST_PHY_SHUTDOWNZ](#), the PHY begins an initialization sequence for normal operation. It is recommended to release [CSI_HOST_PHY_SHUTDOWNZ](#) before [CSI_HOST_DPHY_RSTZ](#). Additionally, ensure that [CSI_DPHY_CFG_CLK](#) (refer to Table 10.2-5 in Chapter 10 [Reset and Clock](#) for details) is available and stable at this point.

The initialization sequence includes several steps, such as enabling internal blocks and performing internal calibrations. Once completed, control transfers to the lanes, which manage power for LP/HS requests from the transmitter by enabling or disabling the corresponding receivers.

Set [CSI_HOST_PHY_STOPSTATEDATA_n](#) and [CSI_HOST_PHY_STOPSTATECLK](#) to 1 to perform all initialization steps. In this state, both clock and data lanes are in stop state (LP-11).

The initialization period (TINIT) is a protocol-dependent parameter with a minimum duration of 100 ns, as defined by the specification. After setting [CSI_HOST_PHY_STOPSTATEDATA_n](#) and [CSI_HOST_PHY_STOPSTATECLK](#) to 1, the software should wait more than 100 ns before starting camera data reception.

40.5.2.4 Control Mode

After completing AFE initialization, the MIPI RX D-PHY remains in Control mode by default. It receives the LP-11 state code (stop state) on the lines until a request is made. Upon receiving a request, the PHY enters either High-Speed Data Reception mode, Escape mode, or Ultra Low Power State. All requests must begin and end with the lanes in the stop state.

40.5.2.5 High-Speed Data Reception Mode

In High-Speed Data Reception mode, the PHY receives data in bursts. The lane operates in high-speed mode only during these bursts.

As shown in Figure 40.5-3, a burst consists of three parts: the low-power initialization sequence, the high-speed data payload, and the end of reception sequence.

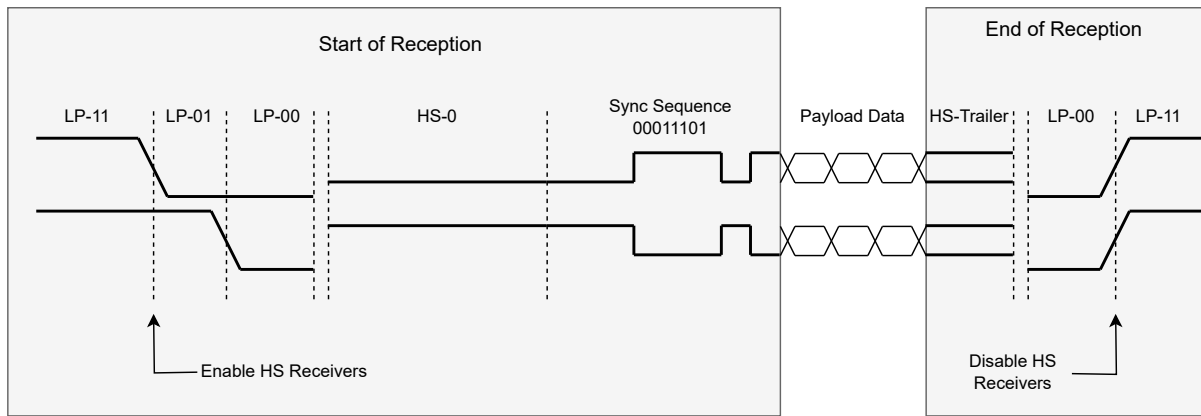


Figure 40.5-3. MIPI CSI HS Data Reception Sequence

The receiver enters high-speed mode following the low-power sequence of [LP-11 > LP-01 > LP-00]. Synchronization is achieved through a sync sequence of “00011101”. After synchronization, the PHY receives high-speed data until a transition from LP-00 to LP-11 is detected on the lane.

40.5.2.6 Escape Mode

Escape mode allows asynchronous communication using data lanes at low speed. During Escape mode, the lane remains in low-power state. A data lane enters this mode after Escape mode request sequence [LP-11 > LP-10 > LP-00 > LP-01 > LP-00]. If an LP-11 state code is detected before the lane reaches LP-00, the request will be aborted, and the lane will return to stop state. Upon entering Escape mode, the transmitter sends an 8-bit command to indicate a requested action.

MIPI CSI only supports Ultra Low Power State (ULPS) in escape mode. The 8-bit command to enter ULPS is “b00011110”. If the entry command is not valid, it will be ignored, and the receiver will wait until the transmitter returns to the stop state.

Ultra Low Power State

Ultra Low Power State involves the lowest power consumption, except for the Shut-Down state.

For data lanes, this mode is activated by sending an Ultra Low Power State entry command “00011110”. After that, the clock lane and data lanes enter the space state (LP-00).

Figure 40.5-4 shows the ULPS sequences for clock and data lanes.

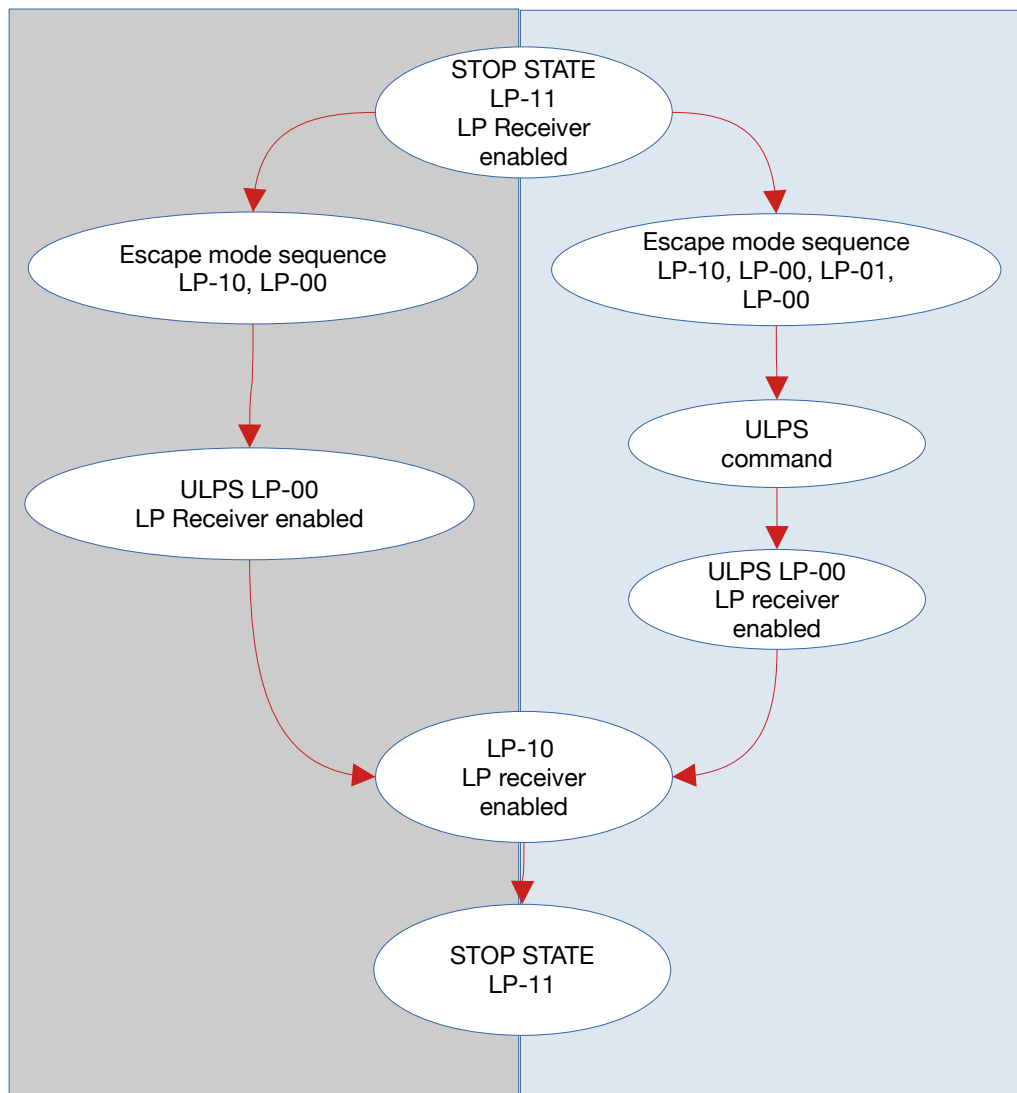


Figure 40.5-4. ULPS Sequences for Clock and Data Lanes

40.5.3 MIPI RX D-PHY Configuration

The MIPI RX D-PHY includes test codes for configuring MIPI RX D-PHY operation. Refer to Section 40.5.3.2 for all supported test codes.

40.5.3.1 MIPI RX D-PHY Programming Interface

The MIPI RX D-PHY test code programming involves two steps: first, program the test code into the registers; second, provide the test data to the tester's inputs.

Before configuring a test code, perform the following steps:

- Write 0 to `CSI_HOST_PHY_SHUTDOWNZ` and `CSI_HOST_DPHY_RSTZ` to shut down and reset the PHY.
- It is also recommended to set `CSI_HOST_PHY_TESTCLR` to 1 to apply a reset pulse before the first configuration of any test code.

Next, configure the test code by following these steps:

- Set the test code.

The test code loads when `CSI_HOST_PHY_TESTEN` is set to 1 and with the falling edge on `CSI_HOST_PHY_TESTCLK`.

- Ensure that `CSI_HOST_PHY_TESTCLK` is set to 1.
- Set `CSI_HOST_PHY_TESTDIN` to the 8-bit test code.
- Set `CSI_HOST_PHY_TESTEN` to 1.
- Set `CSI_HOST_PHY_TESTCLK` to 0. The falling edge on `CSI_HOST_PHY_TESTCLK` latches the value of `CSI_HOST_PHY_TESTDIN[7:0]` as the current test code.
- Set `CSI_HOST_PHY_TESTEN` to 0.
- Enter the test data.

The test data programs to the latest loaded test code when `CSI_HOST_PHY_TESTEN` is 0 and with the rising edge on `CSI_HOST_PHY_TESTCLK`.

- Set `CSI_HOST_PHY_TESTCLK` to 0, if not done already.
- Set `CSI_HOST_PHY_TESTDIN` to the 8-bit test data.
- Set `CSI_HOST_PHY_TESTCLK` to 1. The rising edge on `CSI_HOST_PHY_TESTCLK` programs the test data internally.
- Repeat the above steps to add more test data for the same test code.

Set `CSI_HOST_PHY_TESTCLR` to 1 to reset a test code. This is necessary only before the first programming operation or if you wish to reset the PHY's configuration to its default value.

Figure 40.5-5 shows a timing diagram for the MIPI RX D-PHY control interface. After programming a test code, `CSI_HOST_PHY_TESTDOUT` asynchronously outputs the relevant data associated with the specific test code. This data may include read-back information or other meaningful signals.

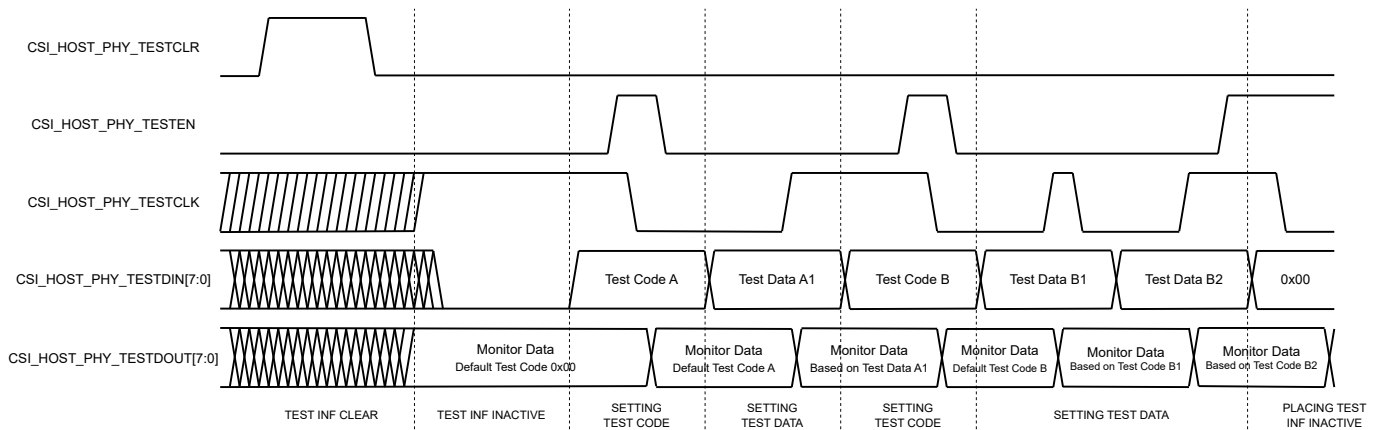


Figure 40.5-5. Control Interface Timing Diagram

Note:

Some test codes require two write data operations. In the first operation, the value of `CSI_HOST_PHY_TESTCLK` changes from 1 to 0. Ensure that the falling edge in the clock does not occur when `CSI_HOST_PHY_TESTEN` is set to 1. Otherwise, the current test data will be latched as an erroneous test code.

It is highly recommended to set the test interface to an inactive state by programming the test code 0x00 when not

used, as shown in the final sequence in Figure 40.5-5.

40.5.3.2 MIPI RX D-PHY Test Code

The MIPI RX D-PHY is configured through test codes via [CSI_HOST_PHY_TEST_CTRL0_REG](#) and [CSI_HOST_PHY_TEST_CTRL1_REG](#). Refer to Section 40.5.3.1 *MIPI RX D-PHY Programming Interface* for more details.

Normal Operation

Use the following test code to enable normal operation of the PHY and the wake-up state of the test interface. In this mode, the test interface is inactive.

- **Test Code:** 0x00
- **Test Data:** None

HS RX Control of Lane 0

Use the following test code to select the HS operating frequency range (hsfreqrange).

- **Test Code:** 0x44
- **Test Data:**

Table 40.5-3. Test Data in HS RX Control of Lane 0

w-1'b0	w-6'b0	w-1'b0
Program Selector	HS operating frequency range selection (hsfreqrange)	Reserved

When Program Selector is 0, Test Data[6:1] programs hsfreqrange.

- **Test Dout:**

Table 40.5-4. Test Dout in HS RX Control of Lane 0

r-1'b0	r-6'b0	r-1'b0
1'b0	HS operating frequency range selection (hsfreqrange)	1'b0

When [CSI_HOST_PHY_TESTDIN](#)[7] is 0, Test Dout[6:1] reflects the value of the programmed hsfreqrange.

Others

Other test codes are reserved.

40.5.4 Descrambler

Data scrambling uses randomization techniques to mitigate electromagnetic interference (EMI) and radio frequency (RF) self-interference. This spreads the transmission energy of the link across a wider frequency band.

The transmitted data is scrambled with a Pseudo-Random-Bit-Stream (PRBS) to reduce the likelihood of repetitive patterns on a lane basis. The PRBS is generated using a Linear Feedback Shift Register (LFSR) that implements the generator polynomial:

$$G(x) = x^{16} + x^5 + x^4 + x^3 + 1 \quad (40.1)$$

Each data lane has an instance of the de-scrambling block. Set [CSI_HOST_SCRAMBLE_ENABLE](#) to 1 to enable this block. Each lane can be configured with a different 16-bit LFSR seed value via [CSI_HOST_SCRAMBLE_SEED_LANE \$n\$](#) .

The default seed values after resetting the controller are listed in [40.5-5](#).

Table 40.5-5. Scrambler PRBS Initial Seed Values for Lane 1 and Lane 2

Lane	Initial Seed Value
1	0x1008
2	0x1188

40.5.5 Error Management

The MIPI CSI-2 host analyzes received packets to identify protocol errors. Possible errors are listed in [Table 40.5-6](#). These errors serve as interrupt sources and can generate the CSI_INTR signal. See [Section 40.6 Interrupts](#) for more details.

Table 40.5-6. Detectable Errors by MIPI CSI-2 Host

Error/Interrupt Source	Description	Corresponding Register Field
PHY_ERRSOTSYNCHS_ n (n : 0-1)	Start-of-transmission error (no synchronization achieved) on data lane n	CSI_HOST_ST_PHY_ERRSOTSYNCHS_n (n: 0-1)
ERR_ECC_DOUBLE	Header ECC contains at least two errors (unrecoverable)	CSI_HOST_ST_ERR_ECC_DOUBLE
SHORTER_PAYLOAD	Reported word count exceeds received word count (unrecoverable)	CSI_HOST_ST_SHORTER_PAYLOAD
ERR_F_BNDRY_MATCH_VC n (n : 0-15)	Error matching frame start with frame end for virtual channel n	CSI_HOST_ST_ERR_F_BNDRY_MATCH_VCn (n: 0-15)
ERR_F_SEQ_VC n (n : 0-15)	Incorrect frame sequence detected on virtual channel n	CSI_HOST_ST_ERR_F_SEQ_VCn (n: 0-15)
ERR_FRAME_DATA_VC n (n : 0-15)	At least one CRC error in the last received frame on virtual channel n	CSI_HOST_ST_ERR_FRAME_DATA_VCn (n: 0-15)
ERR_CRC_VC n (n : 0-15)	Payload CRC error detected on virtual channel n	CSI_HOST_ST_ERR_CRC_VCn (n: 0-15)
ERR_ID_VC n (n : 0-15)	Unrecognized or unimplemented data type detected on virtual channel n	CSI_HOST_ST_ERR_ID_VCn (n: 0-15)

ERR_ECC_CORRECTED_VC n (n : 0-15)	Checksum error detected on virtual channel n	CSI_HOST_ST_ERR_ECC_CORRECTED_VC n (n : 0-15)
PHY_ERRSOTHS_ n (n : 0-1)	Start-of-transmission error (synchronization can still be achieved) on data lane n	CSI_HOST_ST_PHY_ERRSOTHS_ n (n : 0-1)
PHY_ERRESC_ n (n : 0-1)	Escape entry error (ULPS) on data lane n	CSI_HOST_ST_PHY_ERRESC_ n (n : 0-1)

Note:

MIPI CSI does not support virtual channel interleaving. Before starting transmission, verify which virtual channel is in use. Additionally, read error information from the specified virtual channel to detect any errors.

A reset is recommended for all errors in “*_FATAL_REG” registers.

40.6 Interrupts

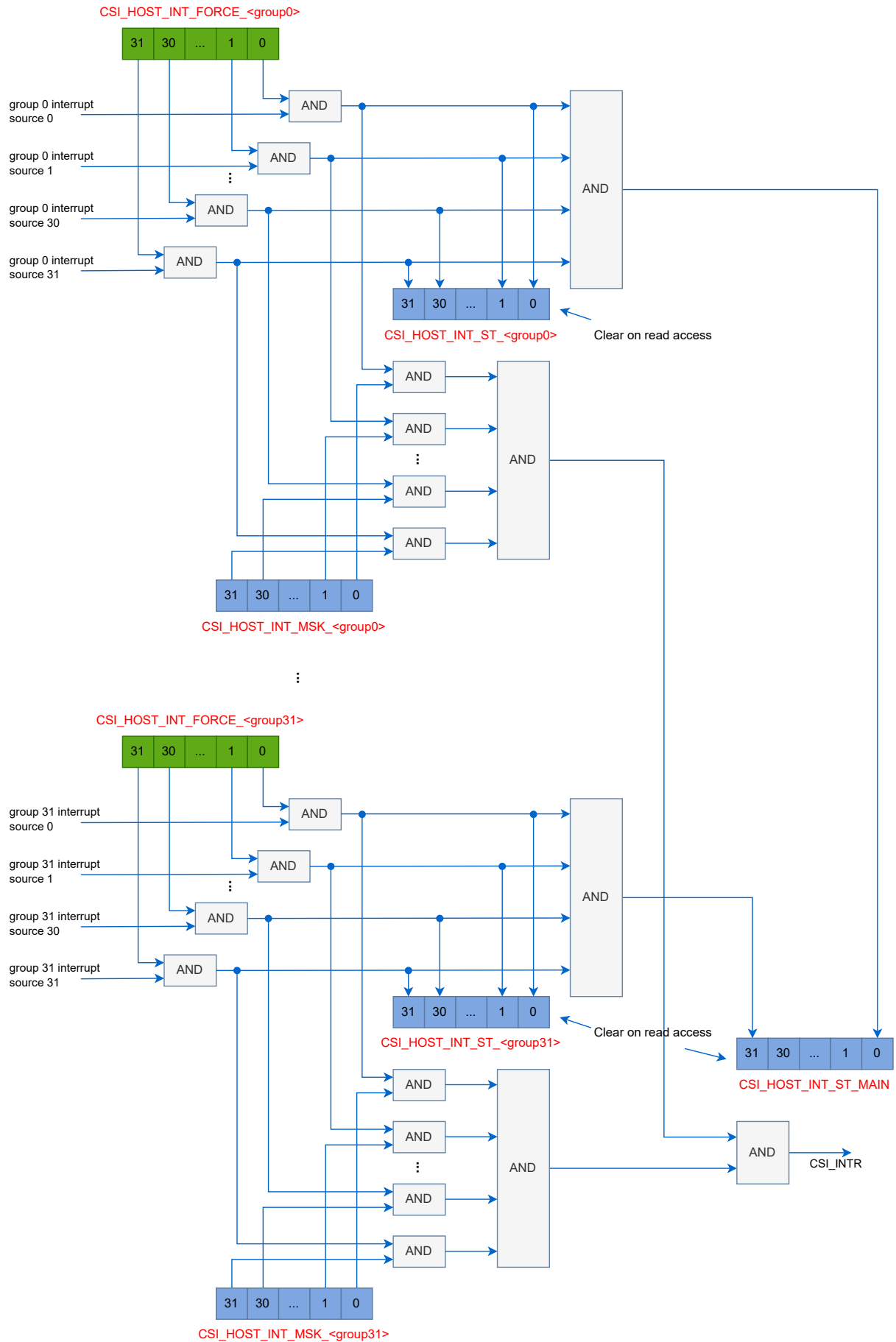
ESP32-P4's MIPI CSI can generate the CSI_INTR interrupt signal that will be sent to the [Interrupt Matrix](#).

The CSI_HOST_INT_ST_<group> registers report error conditions and trigger the CSI_INTR interrupt signal.

You can mask the interrupt pin by using the CSI_HOST_INT_MSK_<group> registers. All errors are masked by default. Set any bit of these registers to 1 to enable the specific error interrupt. The error asserts the respective bit in the CSI_HOST_INT_ST_<group> register. The CSI_HOST_INT_ST_<group> registers clear after a read operation.

Setting any bit of the Interrupt Force registers (CSI_HOST_INT_FORCE_<group>) to 1 triggers the corresponding interrupt without needing to activate the error conditions. These registers are for test purposes.

Figure 40.6-1 shows the interrupt mechanism:



The interrupt signal CSI_INTR can be generated by the following internal interrupt sources (see Table 40.5-6 *Detectable Errors by MIPI CSI-2 Host* for details interrupt source description):

- PHY Fatal Interrupt:
 - PHY_ERRSOTSYNCHS_ *n* (*n*: 0-1)
- Packet Fatal Interrupt:
 - ERR_ECC_DOUBLE
 - SHORTER_PAYLOAD
- Boundary Frame Fatal Interrupt:
 - ERR_F_BNDRY_MATCH_VC*n* (*n*: 0-15)
- Sequence Frame Fatal Interrupt:
 - ERR_F_SEQ_VC*n* (*n*: 0-15)
- CRC Frame Fatal Interrupt:
 - ERR_FRAME_DATA_VC*n* (*n*: 0-15)
- Payload CRC Fatal Interrupt:
 - ERR_CRC_VC*n* (*n*: 0-15)
- Data ID Interrupt:
 - ERR_ID_VC*n* (*n*: 0-15)
- ECC Corrected Interrupt:
 - ERR_ECC_CORRECTED_VC*n* (*n*: 0-15)
- PHY Interrupt:
 - PHY_ERRSOTHS_ *n* (*n*: 0-1)
 - PHY_ERRESC_ *n* (*n*: 0-1)

Note:

For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter 12 *Interrupt Matrix* > Section 12.2 *Interrupt Terminology in ESP32-P4*.

40.7 Programming Procedures

This section describes how to configure the MIPI RX D-PHY and the MIPI CSI-2 host for image data reception. MIPI CSI receives image data only in high-speed data reception mode.

40.7.1 Start High-Speed Data Reception

40.7.1.1 Clock and Reset

- Set `HP_SYS_CLKRST_CSI_HOST_SYS_CLK_EN` to 1 to enable the MIPI CSI clock.

- Set [HP_SYS_CLKRST_RST_EN_CSI_HOST](#) to 0 to release the MIPI CSI reset.
- Configure [HP_SYS_CLKRST_MIPI_CSI_DPHY_CLK_SRC_SEL](#) to select a proper clock source for the MIPI RX D-PHY configuration clock.
- Set [HP_SYS_CLKRST_MIPI_CSI_DPHY_CFG_CLK_EN](#) to 1 to enable the MIPI RX D-PHY configuration clock.
- Set [HP_SYS_CLKRST_RST_EN_CSI_BRG](#) to 1 to release the CSI_Bridge module of the ISP and reset the ISP.

40.7.1.2 MIPI RX D-PHY and MIPI Host Initialization

- Clear [CSI_HOST_CSI2_RESETN](#) to reset the CSI_HOST controller logic.
- Set [CSI_HOST_PHY_TESTCLR](#) to 1, then clear [CSI_HOST_PHY_TESTCLR](#) to reset the test code logic.
- Configure the MIPI RX D-PHY frequency range in test code 0x44 using the MIPI RX D-PHY programming interface.
- Configure [CSI_HOST_N_LANES](#) to define the number of active lanes in the system.
- Set [CSI_HOST_PHY_SHUTDOWNZ](#) and [CSI_HOST_DPHY_RSTZ](#) to 1 to release the MIPI RX D-PHY from the reset state.
- Set the corresponding bit in the [CSI_HOST_INT_MSK_*_REG](#) register to 1 to enable the error interrupt.
- Configure the descrambler function if needed.
- Set [CSI_HOST_CSI2_RESETN](#) to 1 to release the CSI_HOST reset.
- Wait until [CSI_HOST_PHY_STOPSTATEDATA_n](#) and [CSI_HOST_PHY_STOPSTATECLK](#) are 1.
- Start the camera image capture.

40.7.2 Stop High-Speed Data Reception

- Disable related channels in . Wait until is disabled. Refer to Chapter [5 VDMA Controller \(VDMA\)](#).
- Reset the ISP. Refer to Chapter [36 Image Signal Processor \(ISP\)](#).
- Set [HP_SYS_CLKRST_RST_EN_CSI_HOST](#) to 1 to reset the MIPI CSI.

40.8 Register Summary

The addresses in this section are relative to CSI Host base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Version Register			
CSI_HOST_VERSION_REG	CSI_HOST version control register	0x0000	RO
Configuration Registers			
CSI_HOST_N_LANES_REG	Active lane number configuration register	0x0004	R/W
CSI_HOST_CSI2_RESETN_REG	CSI_HOST reset control register	0x0008	R/W
CSI_HOST_PHY_SHUTDOWNZ_REG	RX D-PHY Shut-Down state control register	0x0040	R/W
CSI_HOST_DPHY_RSTZ_REG	RX D-PHY reset control register	0x0044	R/W
CSI_HOST_PHY_RX_REG	RX D-PHY RX-related signal status register	0x0048	RO
CSI_HOST_PHY_TEST_CTRL0_REG	RX D-PHY test code interface control register 0	0x0050	R/W
CSI_HOST_PHY_TEST_CTRL1_REG	RX D-PHY test code interface control register 1	0x0054	varies
CSI_HOST_SCRAMBLING_REG	Descrambler enable register	0x0300	R/W
CSI_HOST_SCRAMBLING_SEED1_REG	Data lane 0 Descrambler seed value control register	0x0304	R/W
CSI_HOST_SCRAMBLING_SEED2_REG	Data lane 1 Descrambler seed value control register	0x0308	R/W
Interrupt Registers			
CSI_HOST_INT_ST_MAIN_REG	Main interrupt status register	0x000C	RC
CSI_HOST_INT_ST_PHY_FATAL_REG	RX D-PHY fatal interrupt status register	0x00E0	RC
CSI_HOST_INT_MSK_PHY_FATAL_REG	RX D-PHY fatal interrupt mask register	0x00E4	R/W
CSI_HOST_INT_FORCE_PHY_FATAL_REG	RX D-PHY fatal interrupt force register	0x00E8	R/W
CSI_HOST_INT_ST_PKT_FATAL_REG	Packet fatal interrupt status register	0x00F0	RC
CSI_HOST_INT_MSK_PKT_FATAL_REG	Packet fatal interrupt mask register	0x00F4	R/W
CSI_HOST_INT_FORCE_PKT_FATAL_REG	Packet fatal interrupt force register	0x00F8	R/W
CSI_HOST_INT_ST_PHY_REG	RX D-PHY interrupt status register	0x0110	RC
CSI_HOST_INT_MSK_PHY_REG	RX D-PHY interrupt mask register	0x0114	R/W
CSI_HOST_INT_FORCE_PHY_REG	RX D-PHY interrupt force register	0x0118	R/W

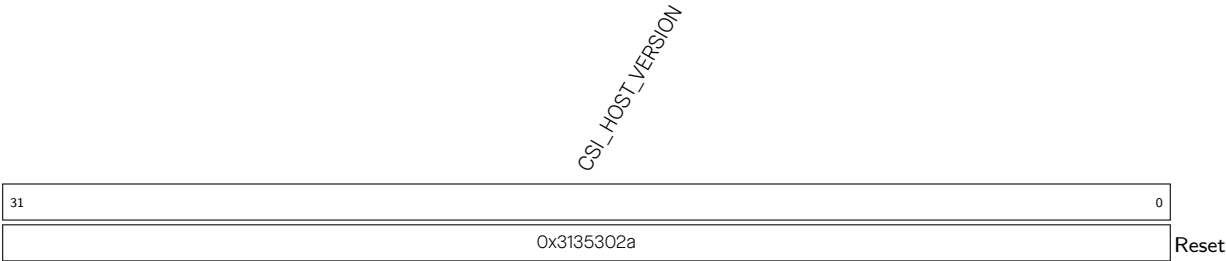
Name	Description	Address	Access
CSI_HOST_INT_ST_BNDRY_FRAME_FATAL_REG	Frame boundary fatal interrupt status register	0x0280	RC
CSI_HOST_INT_MSK_BNDRY_FRAME_FATAL_REG	Frame boundary fatal interrupt mask register	0x0284	R/W
CSI_HOST_INT_FORCE_BNDRY_FRAME_FATAL_REG	Frame boundary fatal interrupt force register	0x0288	R/W
CSI_HOST_INT_ST_SEQ_FRAME_FATAL_REG	Frame sequence fatal interrupt status register	0x0290	RC
CSI_HOST_INT_MSK_SEQ_FRAME_FATAL_REG	Frame sequence fatal interrupt mask register	0x0294	R/W
CSI_HOST_INT_FORCE_SEQ_FRAME_FATAL_REG	Frame sequence fatal interrupt force register	0x0298	R/W
CSI_HOST_INT_ST_CRC_FRAME_FATAL_REG	Frame CRC fatal interrupt status register	0x02A0	RC
CSI_HOST_INT_MSK_CRC_FRAME_FATAL_REG	Frame CRC fatal interrupt mask register	0x02A4	R/W
CSI_HOST_INT_FORCE_CRC_FRAME_FATAL_REG	Frame CRC fatal interrupt force register	0x02A8	R/W
CSI_HOST_INT_ST_PLD_CRC_FATAL_REG	Payload CRC fatal interrupt status register	0x02B0	RC
CSI_HOST_INT_MSK_PLD_CRC_FATAL_REG	Payload CRC fatal interrupt mask register	0x02B4	R/W
CSI_HOST_INT_FORCE_PLD_CRC_FATAL_REG	Payload CRC fatal interrupt force register	0x02B8	R/W
CSI_HOST_INT_ST_DATA_ID_REG	Data ID interrupt status register	0x02C0	RC
CSI_HOST_INT_MSK_DATA_ID_REG	Data ID interrupt mask register	0x02C4	R/W
CSI_HOST_INT_FORCE_DATA_ID_REG	Data ID interrupt force register	0x02C8	R/W
CSI_HOST_INT_ST_ECC_CORRECTED_REG	Header error detected and corrected interrupt status register	0x02D0	RC
CSI_HOST_INT_MSK_ECC_CORRECTED_REG	Header error detected and corrected interrupt mask register	0x02D4	R/W
CSI_HOST_INT_FORCE_ECC_CORRECTED_REG	Header error detected and corrected interrupt force register	0x02D8	R/W
Status Register			
CSI_HOST_PHY_STOPSTATE_REG	RX D-PHY stop state signal status register	0x004C	RO

40.9 Registers

The addresses in this section are relative to CSI Host base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

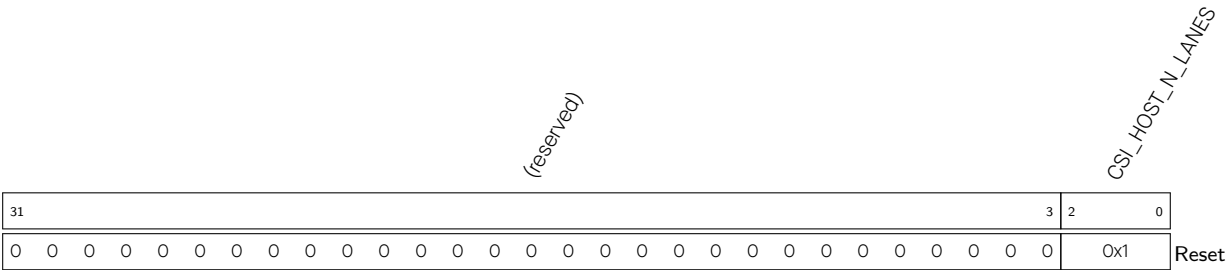
For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

Register 40.1. CSI_HOST_VERSION_REG (0x0000)



CSI_HOST_VERSION Version control register. (RO)

Register 40.2. CSI_HOST_N_LANES_REG (0x0004)

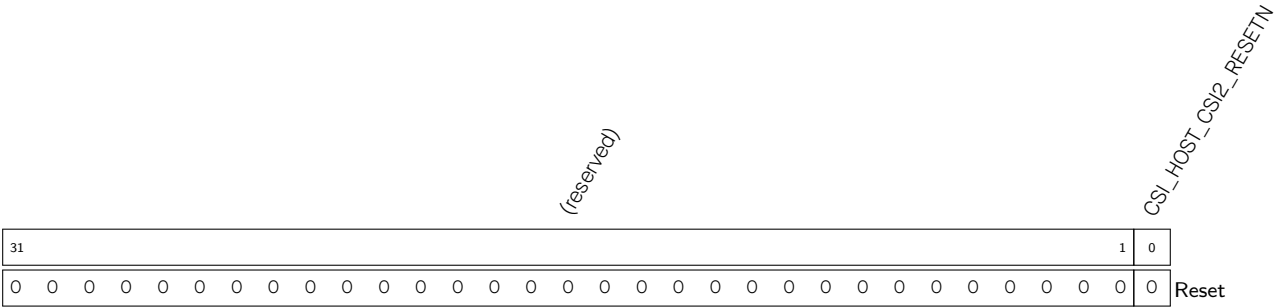


CSI_HOST_N_LANES Configures the number of active lanes that the MIPI CSI-2 host uses to receive the camera device data. This can only be updated when the RX D-PHY lane is in stop state.

- 0: Data lane 0 is active
- 1: Both data lane 0 and data lane 1 are active
- Other values: Invalid

(R/W)

Register 40.3. CSI_HOST_CSI2_RESETN_REG (0x0008)



CSI_HOST_CSI2_RESETN Configures whether to reset the internal logic of the MIPI CSI-2 host controller.

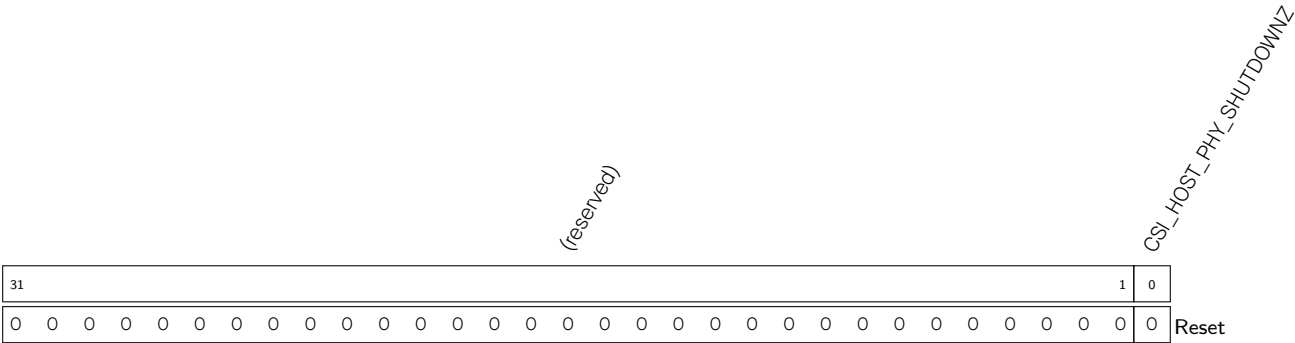
0: Reset

1: Release the reset

This register only affects the internal logic of the controller. It will not reset the configuration to default values.

(R/W)

Register 40.4. CSI_HOST_PHY_SHUTDOWNZ_REG (0x0040)



CSI_HOST_PHY_SHUTDOWNZ Configures whether to enable the Shut-Down state of the RX D-PHY.

0: Enable

1: Disable

(R/W)

Register 40.5. CSI_HOST_DPHY_RSTZ_REG (0x0044)

(reserved)																CSI_HOST_DPHY_RSTZ																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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CSI_HOST_DPHY_RSTZ Configures whether to reset the RX D-PHY digital circuitry.

0: Reset

1: Release the reset

(R/W)

Register 40.6. CSI_HOST_PHY_RX_REG (0x0048)

(reserved)																CSI_HOST_PHY_RXCLKACTIVEHS CSI_HOST_PHY_RXULPSCLKNOT																(reserved)																CSI_HOST_PHY_RXULPSESC_1 CSI_HOST_PHY_RXULPSESC_0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
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CSI_HOST_PHY_RXULPSESC_*n* (*n*: 0-1) Represents whether data lane *n* has entered the Ultra Low Power State.

0: Entered

1: Not entered

(RO)

CSI_HOST_PHY_RXULPSCLKNOT Represents whether RX D-PHY clock lane has entered the Ultra Low Power State.

0: Entered

1: Not entered

(RO)

CSI_HOST_PHY_RXCLKACTIVEHS Represents whether RX D-PHY clock lane is actively receiving a DDR clock.

0: Not actively receiving

1: Actively receiving

(RO)

Register 40.7. CSI_HOST_PHY_TEST_CTRL0_REG (0x0050)

(reserved)																															CSI_HOST_PHY_TESTCLK		CSI_HOST_PHY_TESTCLR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
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0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

CSI_HOST_PHY_TESTCLR Configures whether to initialize the test code interface.

0: Not initialize

1: Initialize

To ensure the analog programmability default values are preset, this bit must be set to 1 then clear to 0 after power-up.

(R/W)

CSI_HOST_PHY_TESTCLK Configures clock to capture [CSI_HOST_PHY_TESTDIN](#) bus contents into D-PHY test code logic, with [CSI_HOST_PHY_TESTEN](#) controlling the operation selection. (R/W)

Register 40.8. CSI_HOST_PHY_TEST_CTRL1_REG (0x0054)

(reserved)																CSL_HOST_PHY_TESTEN		CSL_HOST_PHY_TESTDOUT		CSL_HOST_PHY_TESTDIN						
31															17	16	15				8	7				0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x0				0x0	Reset				

Reset

CSI_HOST_PHY_TESTDIN Configures the 8-bit data input of the test code interface for programming internal registers and accessing test functionalities. (R/W)

CSI_HOST_PHY_TESTDOUT Represents the 8-bit data output of the test code interface for reading data and other probing functionalities. (RO)

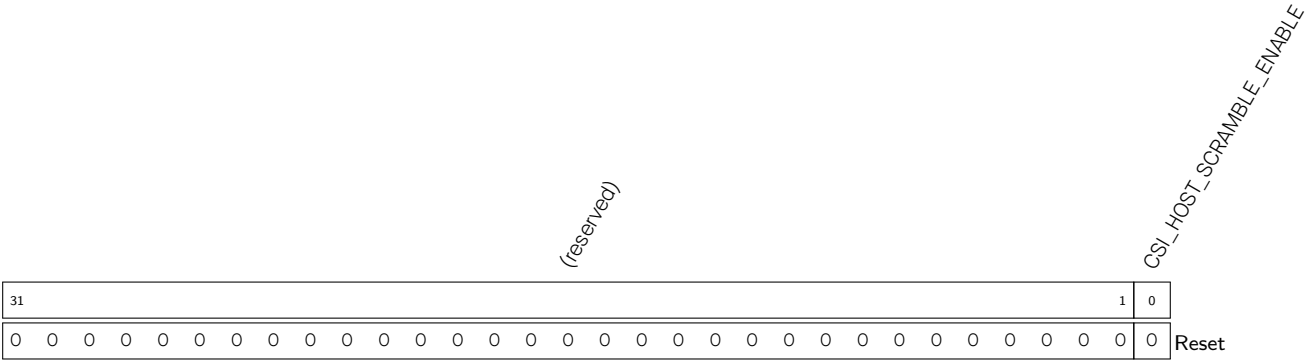
CSI_HOST_PHY_TESTEN Configures whether to performs the test data write operation of the test code interface on the rising edge or the falling edge of [CSI_HOST_PHY_TESTCLK](#).

0: On the rising edge

1: On the falling edge

(R/W)

Register 40.9. CSI_HOST_SCRAMBLING_REG (0x0300)



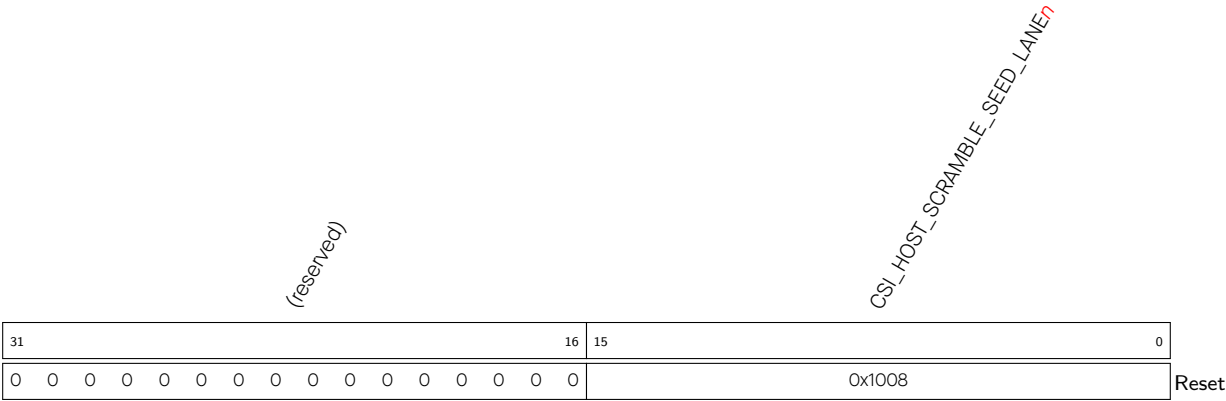
CSI_HOST_SCRAMBLE_ENABLE Configures whether to enable the descrambler.

0: Disable

1: Enable

(R/W)

Register 40.10. CSI_HOST_SCRAMBLING_SEEDn_REG (n: 1-2) (0x0300 + 0x04*n)



CSI_HOST_SCRAMBLE_SEED_LANE_{n-1} Configures the seed value used by the descrambler for lane _{n-1}. (R/W)

Register 40.11. CSI_HOST_INT_ST_MAIN_REG (0x000C)

[illegible]

CSI_HOST_ST_STATUS_INT_PHY_FATAL Represents the status of **CSI_HOST_INT_ST_PHY_FATAL_REG.** (RC)

CSI_HOST_ST_STATUS_INT_PKT_FATAL Represents the status of CSI_HOST_INT_ST_PKT_FATAL_REG. (RC)

CSI_HOST_ST_STATUS_INT_BNDRY_FRAME_FATAL Represents the status of CSI_HOST_INT_ST_BNDRY_FRAME_FATAL_REG. (RC)

CSI_HOST_ST_STATUS_INT_SEQ_FRAME_FATAL Represents the status of CSI_HOST_INT_ST_SEQ_FRAME_FATAL_REG. (RC)

CSI_HOST_ST_STATUS_INT_CRC_FRAME_FATAL Represents the status of CSI_HOST_INT_ST_CRC_FRAME_FATAL_REG. (RC)

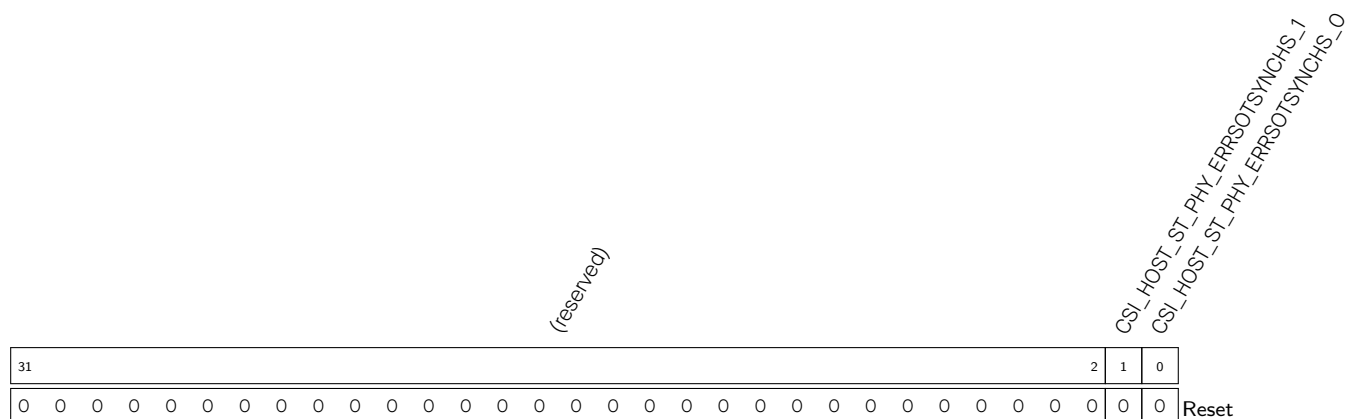
CSI_HOST_ST_STATUS_INT_PLD_CRC_FATAL Represents the status of CSI_HOST_INT_ST_PLD_CRC_FATAL_REG. (RC)

CSI_HOST_ST_STATUS_INT_DATA_ID Represents the status of [CSI_HOST_INT_ST_DATA_ID_REG](#).
(RC)

CSI_HOST_ST_STATUS_INT_ECC_CORRECTED Represents the status of [CSI_HOST_INT_ST_ECC_CORRECTED_REG.](#) (RC)

CSI_HOST_ST_STATUS_INT_PHY Represents the status of **CSI_HOST_INT_PHY_REG**. (RC)

Register 40.12. CSI_HOST_INT_ST_PHY_FATAL_REG (0x00E0)



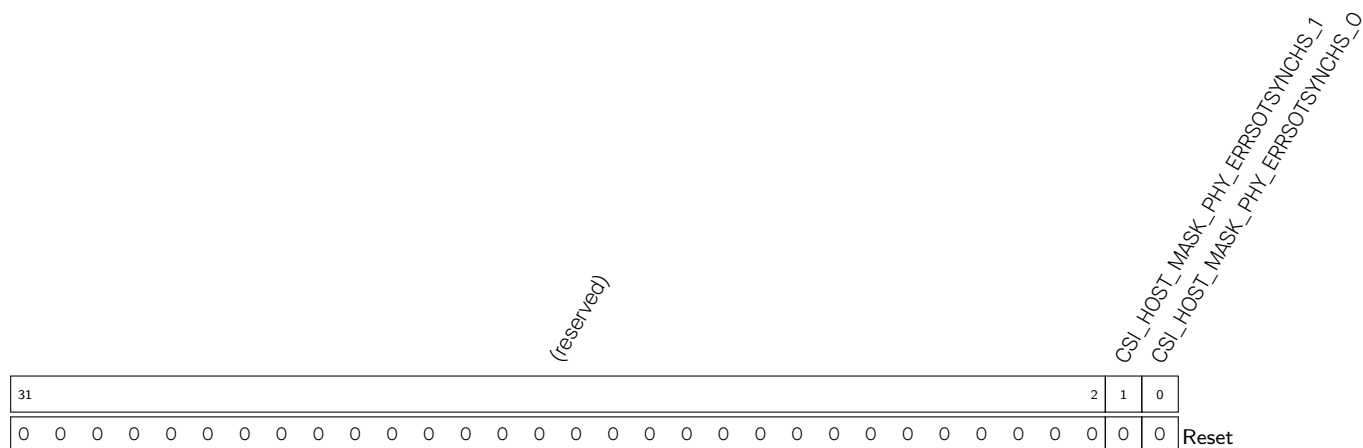
CSI_HOST_ST_PHY_ERRSOTSYNCHS_*n* (*n*: 0-1) Represents whether the PHY_ERRSOTSYNCHS_ error occurs.

0: Do not occur

1: Occur

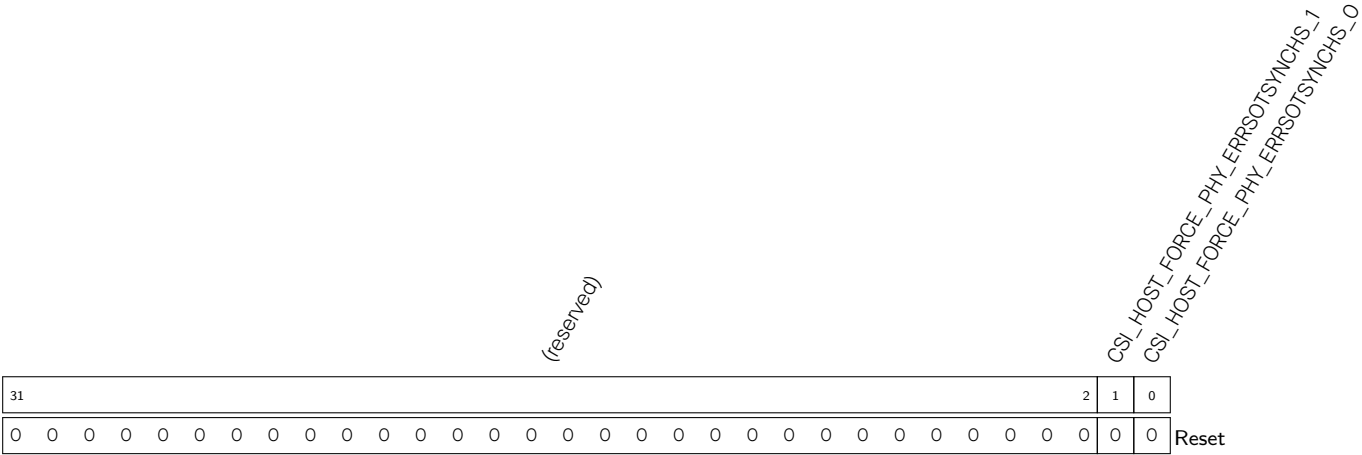
(RC)

Register 40.13. CSI_HOST_INT_MSK_PHY_FATAL_REG (0x00E4)



CSI_HOST_MASK_PHY_ERRSOTSYNCHS_ <i>n</i>	(<i>n</i> : 0-1)	Configures whether to mask
CSI_HOST_ST_PHY_ERRSOTSYNCHS_ <i>n</i> .		
0: Mask the error interrupt		
1: Enable the error interrupt		
(R/W)		

Register 40.14. CSI_HOST_INT_FORCE_PHY_FATAL_REG (0x00E8)



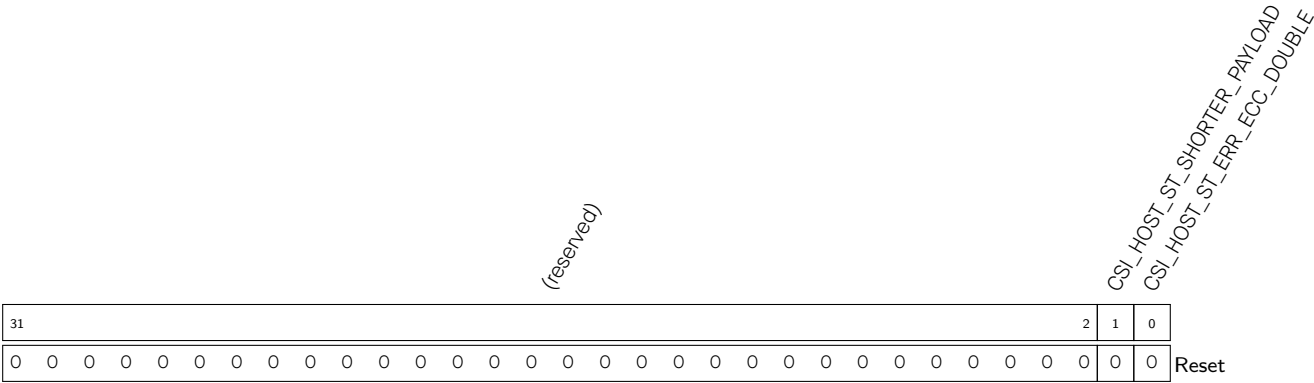
CSI_HOST_FORCE_PHY_ERRSOTSYNCHS_n (*n*: 0-1) Configures whether to force set [CSI_HOST_ST_PHY_ERRSOTSYNCHS_n](#) to 1.

0: Do not force set

1: Force set

(R/W)

Register 40.15. CSI_HOST_INT_ST_PKT_FATAL_REG (0x00F0)



CSI_HOST_ST_ERR_ECC_DOUBLE Represents whether the ERR_ECC_DOUBLE error occurs.

0: Do not occur

1: Occur

(RC)

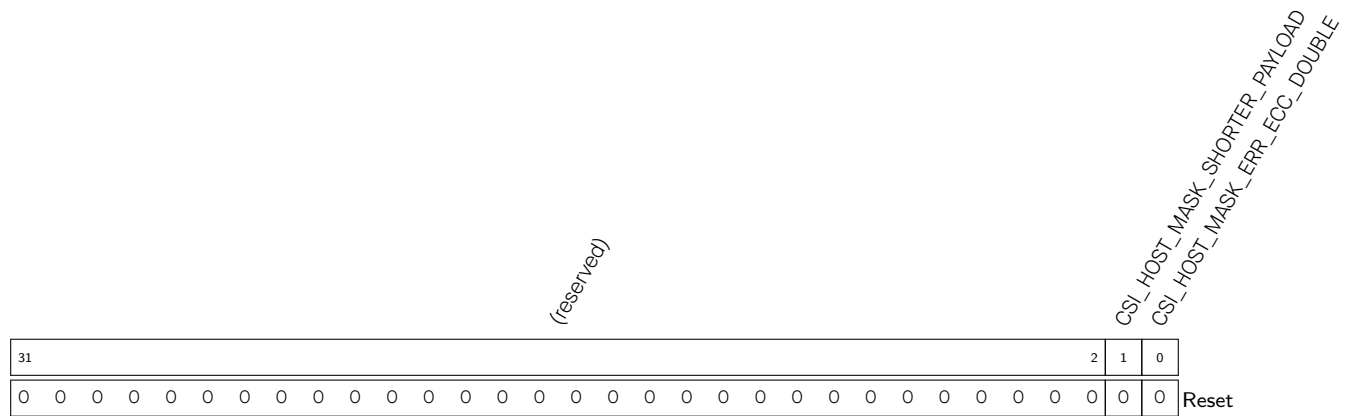
CSI_HOST_ST_SHORTER_PAYLOAD Represents whether the SHORTER_PAYLOAD error occurs.

0: Do not occur

1: Occur

(RC)

Register 40.16. CSI_HOST_INT_MSK_PKT_FATAL_REG (0x00F4)



CSI_HOST_MASK_ERR_ECC_DOUBLE	Configures	whether	to	mask
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CSI_HOST_ST_ERR_ECC_DOUBLE.

0: Mask the error interrupt

1: Enable the error interrupt

(R/W)

CSI_HOST_MASK_SHORTER_PAYLOAD	Configures	whether	to	mask
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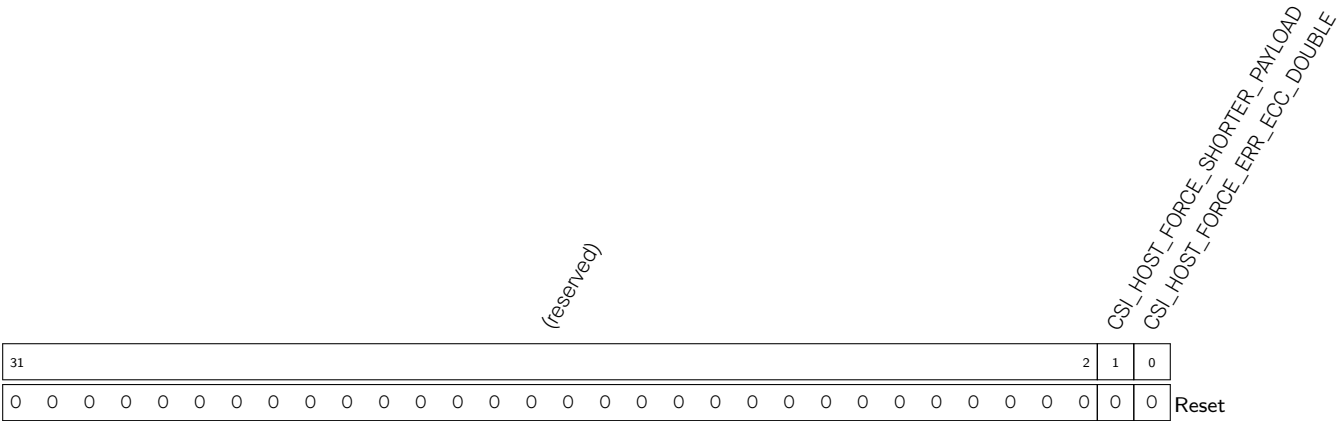
CSI_HOST_ST_SHORTER_PAYLOAD.

0: Mask the error interrupt

1: Enable the error interrupt

(R/W)

Register 40.17. CSI_HOST_INT_FORCE_PKT_FATAL_REG (0x00F8)



CSI_HOST_FORCE_ERR_ECC_DOUBLE

Configures whether to force set CSI_HOST_ST_ERR_ECC_DOUBLE to 1.

0: Do not force set

1: Force set

(R/W)

CSI_HOST_FORCE_SHORTER_PAYLOAD

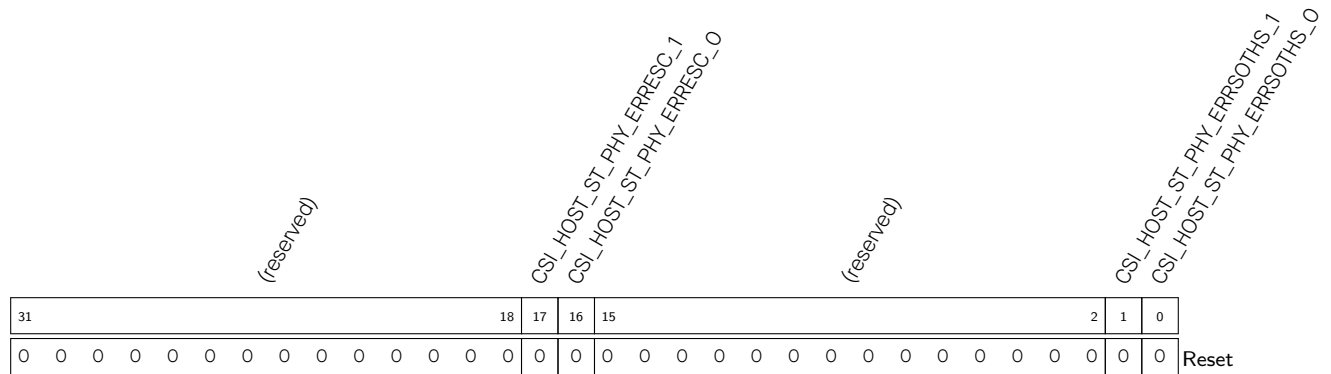
Configures whether to force set CSI_HOST_ST_SHORTER_PAYLOAD to 1.

0: Do not force set

1: Force set

(R/W)

Register 40.18. CSI_HOST_INT_ST_PHY_REG (0x0110)



CSI_HOST_ST_PHY_ERRSOTHS_*n* (*n*: 0-1) Represents whether the PHY_ERRSOTHS_ *n* error occurs.

0: Do not occur

1: Occur

(RC)

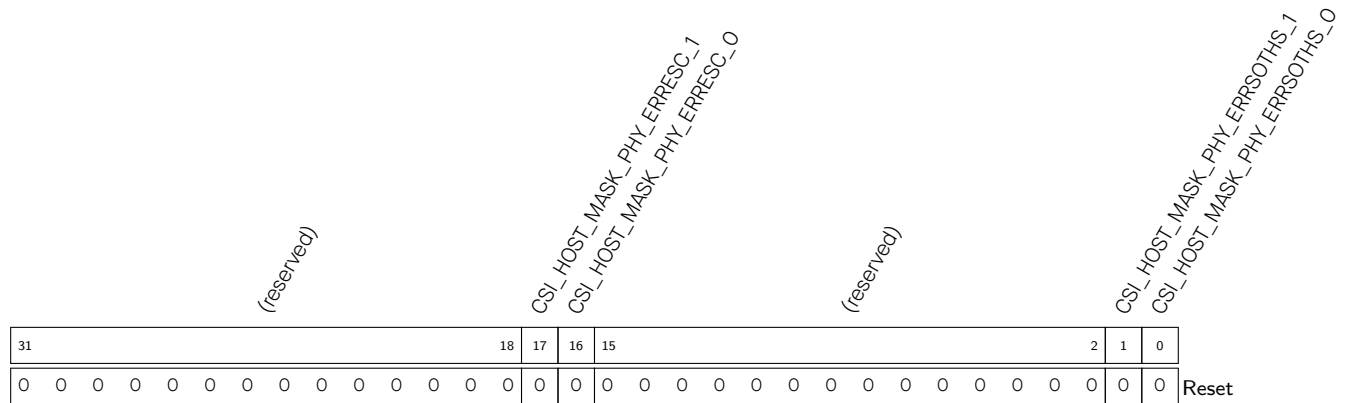
CSI_HOST_ST_PHY_ERRESC_*n* (*n*: 0-1) Represents whether the PHY_ERRESC_*n* error occurs.

0: Do not occur

1: Occur

(RC)

Register 40.19. CSI_HOST_INT_MSK_PHY_REG (0x0114)



CSI_HOST_MASK_PHY_ERRSOTHS_0	Configures whether to mask
CSI_HOST_MASK_PHY_ERRSOTHS_1	CSI_HOST_MASK_PHY_ERRSOTHS_1.
0: Mask the error interrupt	
1: Enable the error interrupt	
(R/W)	

CSI_HOST_MASK_PHY_ERRESC_0 (n: 0-1)	Configures whether to mask CSI_HOST_ST_PHY_ERRESC_0.
0: Mask the error interrupt	
1: Enable the error interrupt	
(R/W)	

Register 40.20. CSI_HOST_INT_FORCE_PHY_REG (0x0118)

[illegible]

CSI_HOST_FORCE_PHY_ERRSOTHS_0 (0-1)	Configures whether to force set CSI_HOST_ST_PHY_ERRSOTHS_0 to 1.
0:	Do not force set
1:	Force set
(R/W)	

CSI_HOST_FORCE_PHY_ERRESC_0	Configures whether to force set CSI_HOST_ST_PHY_ERRESC_0 to 1.	0: Do not force set	1: Force set	(R/W)
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Register 40.21. CSI_HOST_INT_ST_BNDRY_FRAME_FATAL_REG (0x0280)

[illegible]

CSI_HOST_ST_ERR_F_BNDRY_MATCH_VC <i>n</i> (<i>n</i> : 0-15)	Represents	whether	the
ERR_F_BNDRY_MATCH_VC <i>n</i> error occurs.			
0: Do not occur			
1: Occur			
(RC)			

Register 40.22. CSI_HOST_INT_MSK_BNDRY_FRAME_FATAL_REG (0x0284)

(reserved)																CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC15 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC14 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC13 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC12 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC11 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC10 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC9 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC8 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC7 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC6 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC5 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC4 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC3 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC2 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC1 CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC0																
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

CSI_HOST_MASK_ERR_F_BNDRY_MATCH_VC n (n : 0-15) Configures whether to mask
[CSI_HOST_ST_ERR_F_BNDRY_MATCH_VC \$n\$](#) .

0: Mask the error interrupt

1: Enable the error interrupt

(R/W)

Register 40.23. CSI_HOST_INT_FORCE_BNDRY_FRAME_FATAL_REG (0x0288)

(reserved)																CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC15 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC14 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC13 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC12 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC11 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC10 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC9 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC8 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC7 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC6 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC5 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC4 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC3 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC2 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC1 CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC0																
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset							

CSI_HOST_FORCE_ERR_F_BNDRY_MATCH_VC n (n : 0-15) Configures whether to force set
[CSI_HOST_ST_ERR_F_BNDRY_MATCH_VC \$n\$](#) to 1.

0: Do not force set

1: Force set

(R/W)

Register 40.24. CSI_HOST_INT_ST_SEQ_FRAME_FATAL_REG (0x0290)

(reserved)																CSI_HOST_ST_ERR_F_SEQ_VC15 CSI_HOST_ST_ERR_F_SEQ_VC14 CSI_HOST_ST_ERR_F_SEQ_VC13 CSI_HOST_ST_ERR_F_SEQ_VC12 CSI_HOST_ST_ERR_F_SEQ_VC11 CSI_HOST_ST_ERR_F_SEQ_VC10 CSI_HOST_ST_ERR_F_SEQ_VC9 CSI_HOST_ST_ERR_F_SEQ_VC8 CSI_HOST_ST_ERR_F_SEQ_VC7 CSI_HOST_ST_ERR_F_SEQ_VC6 CSI_HOST_ST_ERR_F_SEQ_VC5 CSI_HOST_ST_ERR_F_SEQ_VC4 CSI_HOST_ST_ERR_F_SEQ_VC3 CSI_HOST_ST_ERR_F_SEQ_VC2 CSI_HOST_ST_ERR_F_SEQ_VC1 CSI_HOST_ST_ERR_F_SEQ_VC0																
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset	

CSI_HOST_ST_ERR_F_SEQ_VC n (n : 0-15) Represents whether the ERR_F_SEQ_VC n error occurs.

0: Do not occur

1: Occur

(RC)

Register 40.25. CSI_HOST_INT_MSK_SEQ_FRAME_FATAL_REG (0x0294)

(reserved)																															CSI_HOST_MASK_ERR_F_SEQ_VC15 CSI_HOST_MASK_ERR_F_SEQ_VC14 CSI_HOST_MASK_ERR_F_SEQ_VC13 CSI_HOST_MASK_ERR_F_SEQ_VC12 CSI_HOST_MASK_ERR_F_SEQ_VC11 CSI_HOST_MASK_ERR_F_SEQ_VC10 CSI_HOST_MASK_ERR_F_SEQ_VC9 CSI_HOST_MASK_ERR_F_SEQ_VC8 CSI_HOST_MASK_ERR_F_SEQ_VC7 CSI_HOST_MASK_ERR_F_SEQ_VC6 CSI_HOST_MASK_ERR_F_SEQ_VC5 CSI_HOST_MASK_ERR_F_SEQ_VC4 CSI_HOST_MASK_ERR_F_SEQ_VC3 CSI_HOST_MASK_ERR_F_SEQ_VC2 CSI_HOST_MASK_ERR_F_SEQ_VC1 CSI_HOST_MASK_ERR_F_SEQ_VC0																
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0															
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset													

CSI_HOST_MASK_ERR_F_SEQ_VC n (n : 0-15) Configures whether to mask

CSI_HOST_ST_ERR_F_SEQ_VC n .

0: Mask the error interrupt

1: Enable the error interrupt

(R/W)

Register 40.26. CSI_HOST_INT_FORCE_SEQ_FRAME_FATAL_REG (0x0298)

(reserved)																CSI_HOST_FORCE_ERR_F_SEQ_VC15 CSI_HOST_FORCE_ERR_F_SEQ_VC14 CSI_HOST_FORCE_ERR_F_SEQ_VC13 CSI_HOST_FORCE_ERR_F_SEQ_VC12 CSI_HOST_FORCE_ERR_F_SEQ_VC11 CSI_HOST_FORCE_ERR_F_SEQ_VC10 CSI_HOST_FORCE_ERR_F_SEQ_VC9 CSI_HOST_FORCE_ERR_F_SEQ_VC8 CSI_HOST_FORCE_ERR_F_SEQ_VC7 CSI_HOST_FORCE_ERR_F_SEQ_VC6 CSI_HOST_FORCE_ERR_F_SEQ_VC5 CSI_HOST_FORCE_ERR_F_SEQ_VC4 CSI_HOST_FORCE_ERR_F_SEQ_VC3 CSI_HOST_FORCE_ERR_F_SEQ_VC2 CSI_HOST_FORCE_ERR_F_SEQ_VC1 CSI_HOST_FORCE_ERR_F_SEQ_VC0																
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset							

CSI_HOST_FORCE_ERR_F_SEQ_VC n (n : 0-15) Configures whether to force set CSI_HOST_ST_ERR_F_SEQ_VC n to 1.

0: Do not force set

1: Force set

(R/W)

Register 40.27. CSI_HOST_INT_ST_CRC_FRAME_FATAL_REG (0x02A0)

(reserved)																CSI_HOST_ST_ERR_FRAME_DATA_VC15 CSI_HOST_ST_ERR_FRAME_DATA_VC14 CSI_HOST_ST_ERR_FRAME_DATA_VC13 CSI_HOST_ST_ERR_FRAME_DATA_VC12 CSI_HOST_ST_ERR_FRAME_DATA_VC11 CSI_HOST_ST_ERR_FRAME_DATA_VC10 CSI_HOST_ST_ERR_FRAME_DATA_VC9 CSI_HOST_ST_ERR_FRAME_DATA_VC8 CSI_HOST_ST_ERR_FRAME_DATA_VC7 CSI_HOST_ST_ERR_FRAME_DATA_VC6 CSI_HOST_ST_ERR_FRAME_DATA_VC5 CSI_HOST_ST_ERR_FRAME_DATA_VC4 CSI_HOST_ST_ERR_FRAME_DATA_VC3 CSI_HOST_ST_ERR_FRAME_DATA_VC2 CSI_HOST_ST_ERR_FRAME_DATA_VC1 CSI_HOST_ST_ERR_FRAME_DATA_VC0																	
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

CSI_HOST_ST_ERR_FRAME_DATA_VC n (n : 0-15) Represents whether the ERR_FRAME_DATA_VC n error occurs.

0: Do not occur

1: Occur

(RC)

Register 40.28. CSI_HOST_INT_MSK_CRC_FRAME_FATAL_REG (0x02A4)

(reserved)																CSI_HOST_MASK_ERR_FRAME_DATA_VC15 CSI_HOST_MASK_ERR_FRAME_DATA_VC14 CSI_HOST_MASK_ERR_FRAME_DATA_VC13 CSI_HOST_MASK_ERR_FRAME_DATA_VC12 CSI_HOST_MASK_ERR_FRAME_DATA_VC11 CSI_HOST_MASK_ERR_FRAME_DATA_VC10 CSI_HOST_MASK_ERR_FRAME_DATA_VC9 CSI_HOST_MASK_ERR_FRAME_DATA_VC8 CSI_HOST_MASK_ERR_FRAME_DATA_VC7 CSI_HOST_MASK_ERR_FRAME_DATA_VC6 CSI_HOST_MASK_ERR_FRAME_DATA_VC5 CSI_HOST_MASK_ERR_FRAME_DATA_VC4 CSI_HOST_MASK_ERR_FRAME_DATA_VC3 CSI_HOST_MASK_ERR_FRAME_DATA_VC2 CSI_HOST_MASK_ERR_FRAME_DATA_VC1 CSI_HOST_MASK_ERR_FRAME_DATA_VC0																	
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

CSI_HOST_MSK_ERR_FRAME_DATA_VC n (n : 0-15) Configures whether to mask

CSI_HOST_ST_ERR_FRAME_DATA_VC n .

0: Mask the error interrupt

1: Enable the error interrupt

(R/W)

Register 40.29. CSI_HOST_INT_FORCE_CRC_FRAME_FATAL_REG (0x02A8)

(reserved)																CSI_HOST_FORCE_ERR_FRAME_DATA_VC15 CSI_HOST_FORCE_ERR_FRAME_DATA_VC14 CSI_HOST_FORCE_ERR_FRAME_DATA_VC13 CSI_HOST_FORCE_ERR_FRAME_DATA_VC12 CSI_HOST_FORCE_ERR_FRAME_DATA_VC11 CSI_HOST_FORCE_ERR_FRAME_DATA_VC10 CSI_HOST_FORCE_ERR_FRAME_DATA_VC9 CSI_HOST_FORCE_ERR_FRAME_DATA_VC8 CSI_HOST_FORCE_ERR_FRAME_DATA_VC7 CSI_HOST_FORCE_ERR_FRAME_DATA_VC6 CSI_HOST_FORCE_ERR_FRAME_DATA_VC5 CSI_HOST_FORCE_ERR_FRAME_DATA_VC4 CSI_HOST_FORCE_ERR_FRAME_DATA_VC3 CSI_HOST_FORCE_ERR_FRAME_DATA_VC2 CSI_HOST_FORCE_ERR_FRAME_DATA_VC1 CSI_HOST_FORCE_ERR_FRAME_DATA_VC0																
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset							

CSI_HOST_FORCE_ERR_FRAME_DATA_VC n (n : 0-15) Configures whether to force set

CSI_HOST_ST_ERR_FRAME_DATA_VC n to 1.

0: Do not force set

1: Force set

(R/W)

Register 40.30. CSI_HOST_INT_ST_PLD_CRC_FATAL_REG (0x02B0)

(reserved)																CSI_HOST_ST_ERR_CRC_VC15 CSI_HOST_ST_ERR_CRC_VC14 CSI_HOST_ST_ERR_CRC_VC13 CSI_HOST_ST_ERR_CRC_VC12 CSI_HOST_ST_ERR_CRC_VC11 CSI_HOST_ST_ERR_CRC_VC10 CSI_HOST_ST_ERR_CRC_VC9 CSI_HOST_ST_ERR_CRC_VC8 CSI_HOST_ST_ERR_CRC_VC7 CSI_HOST_ST_ERR_CRC_VC6 CSI_HOST_ST_ERR_CRC_VC5 CSI_HOST_ST_ERR_CRC_VC4 CSI_HOST_ST_ERR_CRC_VC3 CSI_HOST_ST_ERR_CRC_VC2 CSI_HOST_ST_ERR_CRC_VC1 CSI_HOST_ST_ERR_CRC_VC0																	
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

CSI_HOST_ST_ERR_CRC_VC n (n : 0-15) Represents whether the ERR_CRC_VC n error occurs.

0: Do not occur

1: Occur

(RC)

Register 40.31. CSI_HOST_INT_MSK_PLD_CRC_FATAL_REG (0x02B4)

(reserved)																																CSI_HOST_MASK_ERR_CRC_VC15 CSI_HOST_MASK_ERR_CRC_VC14 CSI_HOST_MASK_ERR_CRC_VC13 CSI_HOST_MASK_ERR_CRC_VC12 CSI_HOST_MASK_ERR_CRC_VC11 CSI_HOST_MASK_ERR_CRC_VC10 CSI_HOST_MASK_ERR_CRC_VC9 CSI_HOST_MASK_ERR_CRC_VC8 CSI_HOST_MASK_ERR_CRC_VC7 CSI_HOST_MASK_ERR_CRC_VC6 CSI_HOST_MASK_ERR_CRC_VC5 CSI_HOST_MASK_ERR_CRC_VC4 CSI_HOST_MASK_ERR_CRC_VC3 CSI_HOST_MASK_ERR_CRC_VC2 CSI_HOST_MASK_ERR_CRC_VC1 CSI_HOST_MASK_ERR_CRC_VC0															
31																16																15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset															

CSI_HOST_MASK_ERR_CRC_VC n (n : 0-15) Configures whether to mask

[CSI_HOST_ST_ERR_CRC_VC \$n\$](#) .

0: Mask the error interrupt

1: Enable the error interrupt

(R/W)

Register 40.32. CSI_HOST_INT_FORCE_PLD_CRC_FATAL_REG (0x02B8)

(reserved)																CSI_HOST_FORCE_ERR_CRC_VC15 CSI_HOST_FORCE_ERR_CRC_VC14 CSI_HOST_FORCE_ERR_CRC_VC13 CSI_HOST_FORCE_ERR_CRC_VC12 CSI_HOST_FORCE_ERR_CRC_VC11 CSI_HOST_FORCE_ERR_CRC_VC10 CSI_HOST_FORCE_ERR_CRC_VC9 CSI_HOST_FORCE_ERR_CRC_VC8 CSI_HOST_FORCE_ERR_CRC_VC7 CSI_HOST_FORCE_ERR_CRC_VC6 CSI_HOST_FORCE_ERR_CRC_VC5 CSI_HOST_FORCE_ERR_CRC_VC4 CSI_HOST_FORCE_ERR_CRC_VC3 CSI_HOST_FORCE_ERR_CRC_VC2 CSI_HOST_FORCE_ERR_CRC_VC1 CSI_HOST_FORCE_ERR_CRC_VC0																	
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

CSI_HOST_FORCE_ERR_CRC_VC n (n : 0-15) Configures whether to force set [CSI_HOST_ST_ERR_CRC_VC \$n\$](#) to 1.

0: Do not force set
 1: Force set
 (R/W)

Register 40.33. CSI_HOST_INT_ST_DATA_ID_REG (0x02C0)

(reserved)																CSI_HOST_ST_ERR_ID_VC15 CSI_HOST_ST_ERR_ID_VC14 CSI_HOST_ST_ERR_ID_VC13 CSI_HOST_ST_ERR_ID_VC12 CSI_HOST_ST_ERR_ID_VC11 CSI_HOST_ST_ERR_ID_VC10 CSI_HOST_ST_ERR_ID_VC9 CSI_HOST_ST_ERR_ID_VC8 CSI_HOST_ST_ERR_ID_VC7 CSI_HOST_ST_ERR_ID_VC6 CSI_HOST_ST_ERR_ID_VC5 CSI_HOST_ST_ERR_ID_VC4 CSI_HOST_ST_ERR_ID_VC3 CSI_HOST_ST_ERR_ID_VC2 CSI_HOST_ST_ERR_ID_VC1 CSI_HOST_ST_ERR_ID_VC0																
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset	

CSI_HOST_ST_ERR_ID_VC n (n : 0-15) Represents whether the ERR_ID_VC n error occurs.

0: Do not occur
 1: Occur
 (RC)

Register 40.34. CSI_HOST_INT_MSK_DATA_ID_REG (0x02C4)

(reserved)																CSI_HOST_MASK_ERR_ID_VC15 CSI_HOST_MASK_ERR_ID_VC14 CSI_HOST_MASK_ERR_ID_VC13 CSI_HOST_MASK_ERR_ID_VC12 CSI_HOST_MASK_ERR_ID_VC11 CSI_HOST_MASK_ERR_ID_VC10 CSI_HOST_MASK_ERR_ID_VC9 CSI_HOST_MASK_ERR_ID_VC8 CSI_HOST_MASK_ERR_ID_VC7 CSI_HOST_MASK_ERR_ID_VC6 CSI_HOST_MASK_ERR_ID_VC5 CSI_HOST_MASK_ERR_ID_VC4 CSI_HOST_MASK_ERR_ID_VC3 CSI_HOST_MASK_ERR_ID_VC2 CSI_HOST_MASK_ERR_ID_VC1 CSI_HOST_MASK_ERR_ID_VC0																			
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0			
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

CSI_HOST_MASK_ERR_ID_VC n (n : 0-15) Configures whether to mask [CSI_HOST_ST_ERR_ID_VC \$n\$](#) .

0: Mask the error interrupt

1: Enable the error interrupt

(R/W)

Register 40.35. CSI_HOST_INT_FORCE_DATA_ID_REG (0x02C8)

(reserved)																CSI_HOST_FORCE_ERR_ID_VC15 CSI_HOST_FORCE_ERR_ID_VC14 CSI_HOST_FORCE_ERR_ID_VC13 CSI_HOST_FORCE_ERR_ID_VC12 CSI_HOST_FORCE_ERR_ID_VC11 CSI_HOST_FORCE_ERR_ID_VC10 CSI_HOST_FORCE_ERR_ID_VC9 CSI_HOST_FORCE_ERR_ID_VC8 CSI_HOST_FORCE_ERR_ID_VC7 CSI_HOST_FORCE_ERR_ID_VC6 CSI_HOST_FORCE_ERR_ID_VC5 CSI_HOST_FORCE_ERR_ID_VC4 CSI_HOST_FORCE_ERR_ID_VC3 CSI_HOST_FORCE_ERR_ID_VC2 CSI_HOST_FORCE_ERR_ID_VC1 CSI_HOST_FORCE_ERR_ID_VC0																	
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

CSI_HOST_FORCE_ERR_ID_VC n (n : 0-15) Configures whether to force set [CSI_HOST_ST_ERR_ID_VC \$n\$](#) to 1.

0: Do not force set

1: Force set

(R/W)

Register 40.36. CSI_HOST_INT_ST_ECC_CORRECTED_REG (0x02D0)

(reserved)																CSI_HOST_ST_ERR_ECC_CORRECTED_VC15 CSI_HOST_ST_ERR_ECC_CORRECTED_VC14 CSI_HOST_ST_ERR_ECC_CORRECTED_VC13 CSI_HOST_ST_ERR_ECC_CORRECTED_VC12 CSI_HOST_ST_ERR_ECC_CORRECTED_VC11 CSI_HOST_ST_ERR_ECC_CORRECTED_VC10 CSI_HOST_ST_ERR_ECC_CORRECTED_VC9 CSI_HOST_ST_ERR_ECC_CORRECTED_VC8 CSI_HOST_ST_ERR_ECC_CORRECTED_VC7 CSI_HOST_ST_ERR_ECC_CORRECTED_VC6 CSI_HOST_ST_ERR_ECC_CORRECTED_VC5 CSI_HOST_ST_ERR_ECC_CORRECTED_VC4 CSI_HOST_ST_ERR_ECC_CORRECTED_VC3 CSI_HOST_ST_ERR_ECC_CORRECTED_VC2 CSI_HOST_ST_ERR_ECC_CORRECTED_VC1 CSI_HOST_ST_ERR_ECC_CORRECTED_VC0																	
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

CSI_HOST_ST_ERR_ECC_CORRECTED_VC n (n : 0-15) Represents whether the
 ERR_ECC_CORRECTED_VC n error occurs.

0: Do not occur

1: Occur

(RC)

Register 40.37. CSI_HOST_INT_MSK_ECC_CORRECTED_REG (0x02D4)

(reserved)																CSI_HOST_MASK_ERR_ECC_CORRECTED_VC15 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC14 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC13 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC12 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC11 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC10 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC9 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC8 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC7 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC6 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC5 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC4 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC3 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC2 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC1 CSI_HOST_MASK_ERR_ECC_CORRECTED_VC0																																	
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																	
0																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset															

CSI_HOST_MASK_ERR_ECC_CORRECTED_VC n (n : 0-15) Configures whether to mask
 CSI_HOST_ST_ERR_ECC_CORRECTED_VC n .

0: Mask the error interrupt

1: Enable the error interrupt

(R/W)

Register 40.38. CSI_HOST_INT_FORCE_ECC_CORRECTED_REG (0x02D8)

(reserved)																CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC15 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC14 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC13 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC12 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC11 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC10 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC9 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC8 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC7 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC6 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC5 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC4 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC3 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC2 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC1 CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC0																
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset							

CSI_HOST_FORCE_ERR_ECC_CORRECTED_VC n (n : 0-15) Configures whether to force set **CSI_HOST_ST_ERR_ECC_CORRECTED_VC n** to 1.

0: Do not force set

1: Force set

(R/W)

Register 40.39. CSI_HOST_PHY_STOPSTATE_REG (0x004C)

(reserved)																CSI_HOST_PHY_STOPSTATECLK																(reserved)																CSI_HOST_PHY_STOPSTATEDATA_1																CSI_HOST_PHY_STOPSTATEDATA_0															
31																17																16	15																2																1	0													
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0	0	0	Reset																																											

CSI_HOST_PHY_STOPSTATEDATA_ n (n : 0-1) Represents whether data lane n is in stop state.

0: Not in stop state

1: In stop state

(RO)

CSI_HOST_PHY_STOPSTATECLK Represents whether clock lane in stop state.

0: Not in stop state

1: In stop state

(RO)